Probability and Random Variables

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***Abstract*—This book provides a simple introduction to probability and random variables. The contents are largely based on NCERT textbooks from Class 9-12.**

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1.1. If two events A and B,in a random experiment, $\Pr(A|B) = \frac{4}{5}$ and $\Pr(B|A) = \frac{1}{4}$, then find $\Pr(A|A+B)$.

Solution: From the given information,

$$\frac{\Pr(AB)}{\Pr(B)} = \frac{4}{5}, \tag{1.1.1}$$

$$\frac{\Pr(AB)}{\Pr(A)} = \frac{1}{4} \tag{1.1.2}$$

$$\implies \frac{\Pr(A)}{\Pr(B)} = \frac{5}{16} \tag{1.1.3}$$

$\because$

$$\Pr(A+B) = \Pr(A) + \Pr(B) - \Pr(AB) \tag{1.1.4}$$

$$= \Pr(A) + \frac{5\Pr(A)}{16} - \frac{\Pr(A)}{4} \tag{1.1.5}$$

$$\implies \Pr(A+B) = \frac{17\Pr(A)}{16} \tag{1.1.6}$$

Also

$$\Pr(A|A+B) = \frac{\Pr(A(A+B))}{\Pr(A+B)} \tag{1.1.7}$$

Using Boolean algebra,

$$A(A+B) = A + AB = A(1+B) = A \tag{1.1.8}$$

Thus,

$$\Pr(A|A+B) = \frac{\Pr(A)}{\Pr(A+B)} = \frac{\Pr(A)}{17\Pr(A)/16} \tag{1.1.9}$$

$$= \frac{16}{17} \tag{1.1.10}$$

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